

SymPy Bug Card

Bug 58: Left Limit of `elliptic_k(x)` at 1 Returns `zoo`

Task. Evaluate the real left-hand limit of the complete elliptic integral:

$$\lim_{x \rightarrow 1^-} \text{elliptic_k}(x).$$

SymPy's answer.

$$\text{limit}(\text{elliptic_k}(x), x, 1, \text{dir} = -) = \text{zoo}.$$

Correct answer.

$$\lim_{x \rightarrow 1^-} \text{elliptic_k}(x) = +\infty.$$

Explanation. For real $m \rightarrow 1^-$, $K(m)$ has leading behavior

$$K(m) = -\frac{1}{2} \log(1 - m) + O(1),$$

so it diverges through positive real values. Returning `zoo` loses the one-sided real sign and is not equivalent to $+\infty$.

Likely root cause. The diagnosis traces the limit through `elliptic_k._eval_nseries`, which delegates the expansion near $m = 1$ to a hypergeometric rewrite that returns `nan` for `elliptic_k(1 - w)`. The missing branch-aware logarithmic expansion makes the limit machinery fall back to unsigned complex infinity; this is a new instance of a known failure mode.

Artifact (GitHub).

[bug-report/bug-058-left-limit-of-elliptic-k-x-at-1-returns-zoo/](https://github.com/sympy/sympy/issues/15111)